ThreeDotProgressIndicator Documentation (Flutter)

Overview

`ThreeDotProgressIndicator` is a **custom animated loading widget** built using Flutter. It displays a horizontal row of dots where one dot becomes active at a time, creating a looping animation.

Common Use Cases

- Loading indicators
- Chat typing animation
- Waiting states
- Onboarding or step-based UI feedback

This widget is **lightweight**, **reusable**, and **fully customizable**.

What This Widget Does

- Displays multiple circular dots
- Animates the active dot in a loop
- Uses a `Timer` to control animation steps
- Smooth color transitions using `AnimatedContainer`

Full Source Code

```
import 'package:flutter/material.dart';
import 'dart:async';

class ThreeDotProgressIndicator extends StatefulWidget {
  /// Number of dots to display
  final int dotCount;

  /// Size of each dot (width & height)
  final double dotSize;

  /// Space between dots
  final double spacing;

  /// Color of the active dot
  final Color activeColor;
```

```dart
  /// Color of inactive dots
  final Color inactiveColor;

  /// Duration between dot transitions
  final Duration duration;

  const ThreeDotProgressIndicator({
    Key? key,
    this.dotCount = 3,
    this.dotSize = 8,
    this.spacing = 8,
    this.activeColor = Colors.blue,
    this.inactiveColor = Colors.grey,
    this.duration = const Duration(milliseconds: 600),
  }) : super(key: key);

  @override
  State<ThreeDotProgressIndicator> createState() =>
      _ThreeDotProgressIndicatorState();
}

class _ThreeDotProgressIndicatorState extends State<ThreeDotProgressIndicator> {
  int _currentStep = 0;
  Timer? _timer;

  @override
  void initState() {
    super.initState();

    _timer = Timer.periodic(widget.duration, (_) {
      if (!mounted) return;
      setState(() {
        _currentStep = (_currentStep + 1) % widget.dotCount;
      });
    });
  }

  @override
  void dispose() {
    _timer?.cancel();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return Row(
      mainAxisSize: MainAxisSize.min,
```

```
      children: List.generate(
        widget.dotCount,
        (index) => AnimatedContainer(
          duration: const Duration(milliseconds: 300),
          margin: EdgeInsets.symmetric(horizontal: widget.spacing / 2),
          width: widget.dotSize,
          height: widget.dotSize,
          decoration: BoxDecoration(
            shape: BoxShape.circle,
            color: index == _currentStep
                ? widget.activeColor
                : widget.inactiveColor,
          ),
        ),
      ),
    );
  }
}
```

Logic Explanation

State Variables

- `_currentStep` → Tracks which dot is active
- `_timer` → Controls the animation loop

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initState()

- Starts a periodic timer
- Updates `_currentStep` every `duration`
- Uses modulo `% dotCount` for infinite looping

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dispose()

- Cancels the timer
- Prevents memory leaks

UI Explanation

Row Widget

- Displays dots horizontally
- `mainAxisSize.min` keeps layout compact

AnimatedContainer

- Animates color changes smoothly
- Automatically animates when properties change

Property Breakdown

dotCount

Number of dots displayed

dotSize

Controls dot width and height

spacing

Space between dots

activeColor

Color of the active dot

inactiveColor

Color of inactive dots

duration

Speed of the animation loop

Example Usage

```
ThreeDotProgressIndicator(
  dotCount: 3,
  dotSize: 10,
```

```
  spacing: 12,
  activeColor: Colors.green,
  inactiveColor: Colors.grey.shade400,
  duration: Duration(milliseconds: 500),
)
```

Best Practices

- Always cancel `Timer` in `dispose()`
- Keep animation durations subtle
- Use soft colors for loaders
- Ideal for chat & loading UI

Summary

`ThreeDotProgressIndicator` is a clean, efficient, and reusable Flutter widget for loading animations. It improves user experience while keeping performance overhead minimal.